SIMATS ENGINEERING
ASSESSMENT TOOL 3: Comparative Analysis

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage: 10%

QUESTIONS: 5 **Total Marks:50**

Annexure A

Comparative Analysis

Code of Conduct:

I **M. Phanith Chandu (192425238)** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: M. Phanith Chandu

Q. No	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)	Total Score (100%)
1						
2						
3						
4						
5						
Total Marks (Out of 100)				Total Marks (Out of 50)		

Annexure A

Comparative Analysis

Compute the required performance measures from the given data and compare the techniques based on the results. Evaluate and justify your conclusion using the calculated values.

1. *Detection Performance Comparison*

Two object detection models produce the following results:

Model True Positives False Positives False Negatives

A (Viola-Jones) 80 20 30

B (YOLO) 90 15 20

(a) Compute precision and recall for both models.

(b) Compare their performance and state which model is better for object detection.

2. *Motion Estimation Error Analysis*

Two motion estimation techniques produce displacement errors (in pixels):

Frame Method A Method B

1 2 3

2 3 5

3 2 4

(a) Compute average error for both methods.

(b) Compare and evaluate which method is more accurate.

3. *Optical Flow Magnitude Comparison*

Optical flow magnitudes (in pixels/frame):

Region Method A Method B

R1 5 8

R2 6 10

R3 5 9

(a) Compute average motion magnitude for both methods.

(b) Compare and evaluate which method captures stronger motion.

4. Pattern Recognition Accuracy Comparison

Two recognition systems produce results:

System Correct Predictions Total Samples

Traditional 85 100

Deep Learning 92 100

(a) Compute accuracy for both systems.
(b) Compare and evaluate which system performs better.
5. *Detection Accuracy Comparison*
Results:

Model	*Correct*	*Total*
A	180	200
B	170	200

(a) Compute accuracy.
(b) Compare and evaluate.

1. Detection Performance Comparison
Given data:

Model	True Positives	False Positives	False Negatives
A (Viola-Jones)	80	20	30
B (YOLO)	90	15	20

(a) Precision and Recall

Precision = TP / (TP + FP)

Recall = TP / (TP + FN)

Model A (Viola-Jones):

- Precision = 80 / (80 + 20) = 80 / 100 = 0.80 = 80%.
- Recall = 80 / (80 + 30) = 80 / 110 ≈ 72.73%.

Model B (YOLO):

- Precision = 90 / (90 + 15) = 90 / 105 ≈ 85.71%.
- Recall = 90 / (90 + 20) = 90 / 110 ≈ 81.82%.

(b) *Comparison and Evaluation*

- YOLO has higher precision: 85.71% compared with 80% for Viola-Jones.
- YOLO has higher recall: 81.82% compared with 72.73% for Viola-Jones.
- YOLO produces fewer false positives (15 vs 20) and fewer false negatives (20 vs 30).
- Higher precision means YOLO's positive detections are more likely to be correct.
- Higher recall means YOLO detects a larger proportion of the actual objects.
- Therefore, YOLO performs better on both important detection measures.

Critical Interpretation

YOLO is the stronger model because it improves both precision and recall rather than improving only one measure. The lower false-positive and false-negative counts provide direct evidence for this decision. For an object-detection application where both reliable detections and fewer missed objects matter, YOLO is preferable.

2. Motion Estimation Error Analysis

Given displacement errors:

Frame	Method A	Method B
1	2	3
2	3	5
3	2	4

(a) *Average Error*

Average error = Sum of frame errors / Number of frames

- Method A = (2 + 3 + 2) / 3 = 7 / 3 ≈ 2.33 pixels.
- Method B = (3 + 5 + 4) / 3 = 12 / 3 = 4 pixels.

(b) *Comparison and Evaluation*

- Method A has an average error of approximately 2.33 pixels.
- Method B has an average error of 4 pixels.
- Method A therefore has approximately 1.67 pixels less average error.
- Method A also has lower error in every individual frame: 2 vs 3, 3 vs 5, and 2 vs 4.
- Lower displacement error indicates more accurate motion estimation.

Critical Interpretation

Method A is more accurate because its average error is substantially lower and it performs better in every listed frame. Based strictly on the supplied error data, Method A should be selected when accurate displacement estimation is the main objective.

3. Optical Flow Magnitude Comparison

Given optical flow magnitudes:

Region	Method A	Method B
R1	5	8
R2	6	10
R3	5	9

(a) *Average Motion Magnitude*

Average magnitude = Sum of magnitudes / Number of regions

- Method A = (5 + 6 + 5) / 3 = 16 / 3 ≈ 5.33 pixels/frame.
- Method B = (8 + 10 + 9) / 3 = 27 / 3 = 9 pixels/frame.

(b) *Comparison and Evaluation*

- Method A has an average motion magnitude of approximately 5.33 pixels/frame.
- Method B has an average of 9 pixels/frame.
- Method B captures approximately 3.67 pixels/frame more average motion magnitude.
- Method B is higher in every region: 8 vs 5 in R1, 10 vs 6 in R2, and 9 vs 5 in R3.
- Therefore, Method B captures stronger motion according to the supplied magnitude values.

Critical Interpretation

Method B is the stronger method for capturing larger motion magnitude in the given regions. However, a higher magnitude does not automatically mean higher accuracy; it only shows that the measured motion is stronger according to the provided data. The conclusion is therefore specifically about motion magnitude, not overall motion-estimation quality.

4. Pattern Recognition Accuracy Comparison

Given data:

System	Correct Predictions	Total Samples
Traditional	85	100
Deep Learning	92	100

(a) *Accuracy Calculation*

Accuracy = Correct Predictions / Total Samples × 100

- Traditional = 85 / 100 × 100 = 85%.
- Deep Learning = 92 / 100 × 100 = 92%.

(b) *Comparison and Evaluation*

- Deep Learning has 92% accuracy, while Traditional recognition has 85%.
- Deep Learning is 7 percentage points more accurate.
- Out of 100 samples, Deep Learning correctly recognizes 7 more samples.
- Both systems are evaluated on the same number of samples, so the comparison is direct.
- Based on the supplied accuracy data, Deep Learning performs better.

Critical Interpretation

Deep Learning is preferable when recognition accuracy is the main requirement. The result shows a clear performance advantage of 7 percentage points. The supplied data does not provide processing time, computational cost, or training requirements, so those factors cannot be used to claim that Deep Learning is better in every practical situation.

5. Detection Accuracy Comparison

Given data:

Model	Correct	Total
A	180	200
B	170	200

(a) *Accuracy Calculation*

Accuracy = Correct Predictions / Total Samples × 100

- Model A = 180 / 200 × 100 = 90%.
- Model B = 170 / 200 × 100 = 85%.

(b) *Comparison and Evaluation*

- Model A has 90% accuracy, while Model B has 85%.
- Model A is 5 percentage points more accurate.
- Model A correctly detects 10 more samples than Model B out of the same 200 samples.
- Because both models use the same total number of samples, the accuracy comparison is straightforward.
- Model A performs better according to the supplied accuracy measure.

Critical Interpretation

Model A should be selected if detection accuracy is the primary evaluation criterion. However, the supplied data contains only correct and total counts, so other factors such as latency, computational cost, robustness, and false-positive behavior cannot be evaluated from this dataset.